Antenna Test Report

Dual-band 2.4/5 GHz whip (simulated)

Measured 2026-10-11 14:00 CEST · Report generated 2026-10-12 10:02 CEST · AntennaFox 1.0.0

Report AFX-20261011-SAMPLE

SIMULATION – NOT A REAL MEASUREMENT

Well matched in all measured target bands

The antenna accepts the transmitter power across the whole target band. How much of it is radiated is a separate question – see the gain test.

Scope: one-port S11 at the reference plane 'Unknown'; antenna not stated; acceptance S11 ≤ -10 dB (VSWR ≤ 1.92) at every point of each target band; no extrapolation beyond the measured span.

Worst target band	Wi-Fi 5 GHz full range (5150–5850 MHz) – Well matched (worst S11 -13.7 dB)
Evidence quality	SIMULATION – not a real measurement
Calibration	Not verified
Gain	Not measured
Radiation pattern	Not measured
Measured span	2000.000–6000.000 MHz
Instrument	Simulation – no instrument

⚠ Frequency match does not prove performance, gain, protocol operation, or regulatory suitability.

⚠ Band compatibility ≠ protocol/device compatibility: an S11 measurement cannot validate, for example, antenna certification for a radio module, antenna diversity, transmit power limits, or link range.

FREQUENCY / PROTOCOL BANDS

Band	Region	Coverage	Worst S11	Power accepted	Max VSWR	Span ≤ -10 dB	Pts	Result
ISM 2.4 GHz (regulatory band)	global	Measured (100%)	-14.4 dB	≥ 96 % (≤ 0.2 dB)	1.47	100%	84	**Well matched**
Wi-Fi 2.4 GHz channels 1–13 (EU, 20 MHz)	EU	Measured (100%)	-14.6 dB	≥ 96 % (≤ 0.2 dB)	1.46	100%	81	**Well matched**
Bluetooth LE (40 × 2 MHz channels)	global	Measured (100%)	-14.5 dB	≥ 96 % (≤ 0.2 dB)	1.46	100%	81	**Well matched**
Wi-Fi 5 GHz full range (5150–5850 MHz)	global	Measured (100%)	-13.7 dB	≥ 95 % (≤ 0.2 dB)	1.52	100%	701	**Well matched**
Wi-Fi 5 GHz EU band A (5150–5250 MHz)	EU	Measured (100%)	-23.4 dB	≥ 99 % (≤ 0.1 dB)	1.14	100%	101	**Well matched**
Wi-Fi 5 GHz EU band A2 (5250–5350 MHz, DFS)	EU	Measured (100%)	-29.6 dB	≥ 99 % (≤ 0.1 dB)	1.07	100%	101	**Well matched**
Wi-Fi 5 GHz EU band B (5470–5725 MHz, DFS)	EU	Measured (100%)	-15.9 dB	≥ 97 % (≤ 0.2 dB)	1.38	100%	256	**Well matched**
Wi-Fi 5 GHz U-NII-3 (5725–5850 MHz, US)	US	Measured (100%)	-13.7 dB	≥ 95 % (≤ 0.2 dB)	1.52	100%	126	**Well matched**

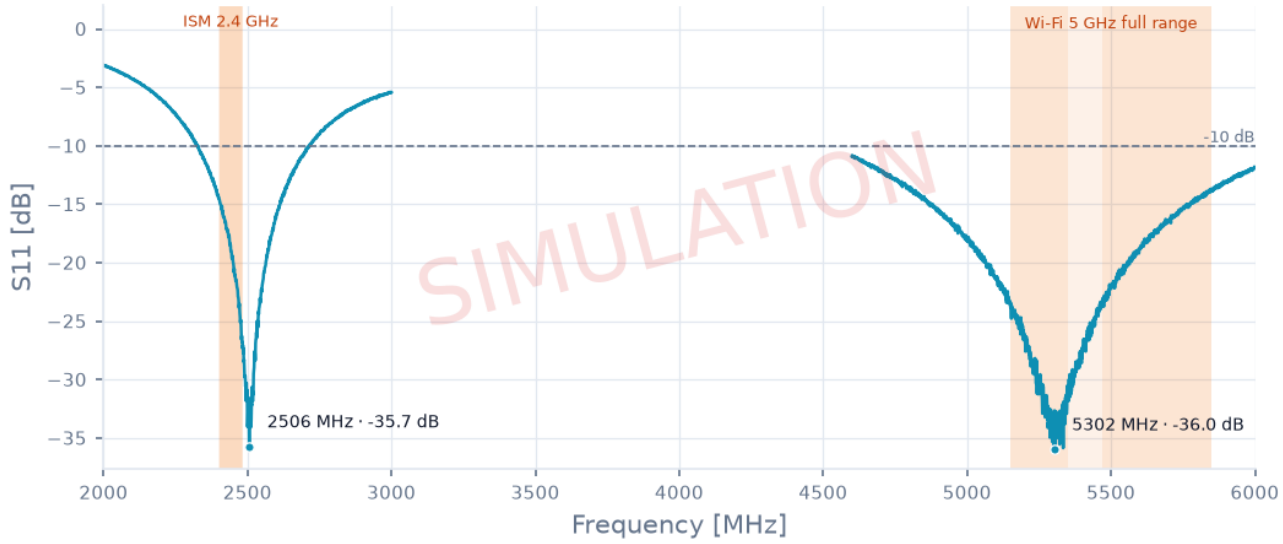
Thresholds: well matched ≤ -10 dB (VSWR ≈ 1.92), marginal ≤ -6 dB. Each band is rated by its worst value over the entire required interval; no extrapolation beyond the measured span. 'Power accepted' is the share of the incident power not reflected at the worst point ($1 - |\Gamma|^2$), rounded down, with the mismatch loss rounded up; it says nothing about how much is radiated. 'Span ≤ -10 dB' is the share of the band's frequency span that lies below the threshold. Convention: S11 [dB] = 20·log10|Γ| (negative = matched), return loss RL = −S11.

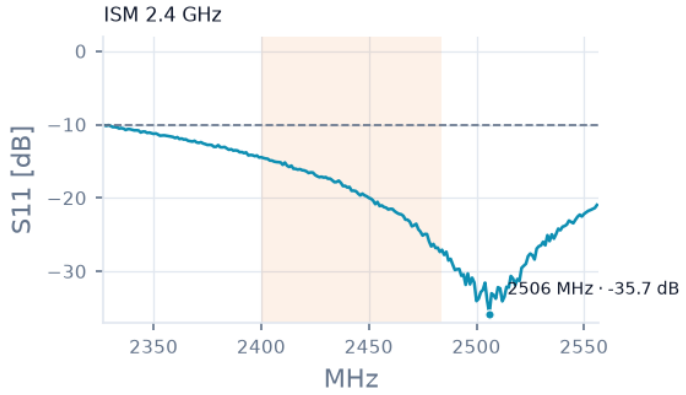
At band centre

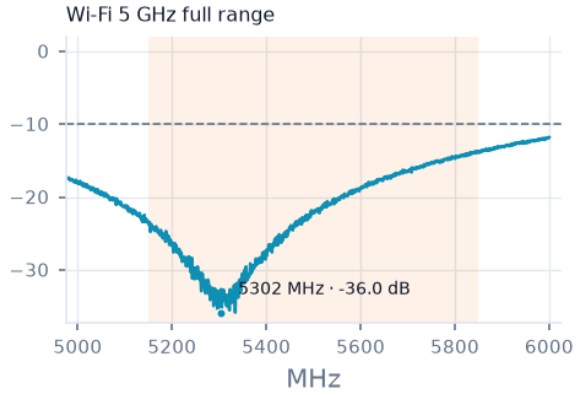
Band	Centre	S11	VSWR	Impedance	Power accepted
ISM 2.4 GHz (regulatory band)	2442.000 MHz	-19.0 dB	1.25	43.8 − j8.6 Ω	≥ 98 % (≤ 0.1 dB)
Wi-Fi 2.4 GHz channels 1–13 (EU, 20 MHz)	2442.000 MHz	-19.0 dB	1.25	43.8 − j8.6 Ω	≥ 98 % (≤ 0.1 dB)
Bluetooth LE (40 × 2 MHz channels)	2441.000 MHz	-18.5 dB	1.27	43.6 − j9.2 Ω	≥ 98 % (≤ 0.1 dB)
Wi-Fi 5 GHz full range (5150–5850 MHz)	5500.000 MHz	-22.0 dB	1.17	50.1 + j8.0 Ω	≥ 99 % (≤ 0.1 dB)

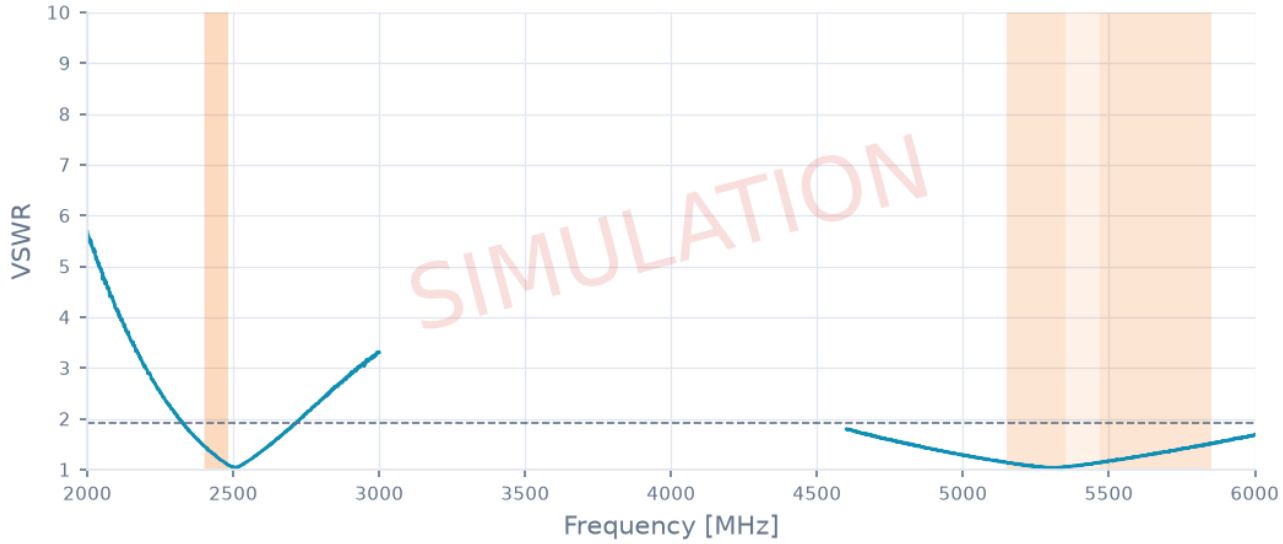
Band	Centre	S11	VSWR	Impedance	Power accepted
Wi-Fi 5 GHz EU band A (5150–5250 MHz)	5200.000 MHz	-26.4 dB	1.10	52.9 − j4.0 Ω	≥ 99 % (≤ 0.1 dB)
Wi-Fi 5 GHz EU band A2 (5250–5350 MHz, DFS)	5300.000 MHz	-32.9 dB	1.05	52.3 − j0.3 Ω	≥ 99 % (≤ 0.1 dB)
Wi-Fi 5 GHz EU band B (5470–5725 MHz, DFS)	5597.000 MHz	-19.2 dB	1.25	49.5 + j11.0 Ω	≥ 98 % (≤ 0.1 dB)
Wi-Fi 5 GHz U-NII-3 (5725–5850 MHz, US)	5787.000 MHz	-14.9 dB	1.44	48.2 + j17.9 Ω	≥ 96 % (≤ 0.2 dB)

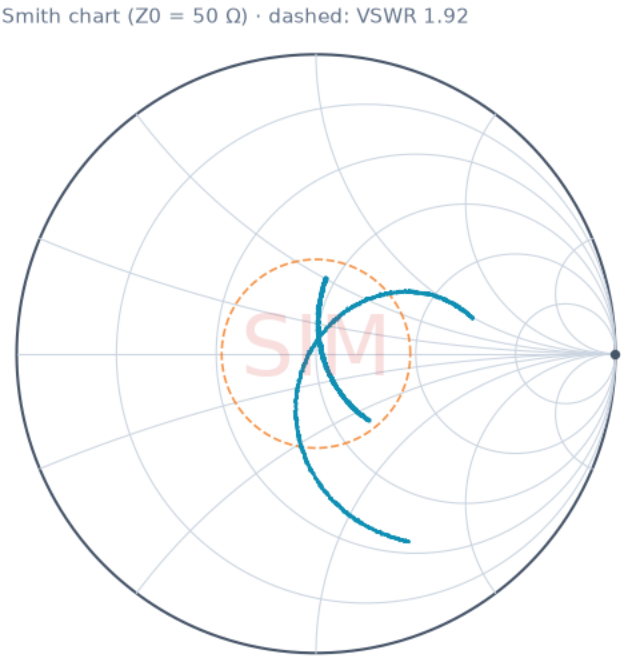
MEASURED TRACES











KEY VALUES

Minimum S11	-36.0 dB at 5302.000 MHz
Impedance at minimum	51.6 + j0.2 Ω, VSWR 1.03
Windows S11 ≤ -10 dB	2323.9–2325.1 MHz (1.1 MHz), 2326.5–2716.1 MHz (389.6 MHz), 4600.0–6000.0 MHz (1400.0 MHz)
Resonances (minima)	2506.0 MHz (-35.7 dB), 5302.0 MHz (-36.0 dB), 5305.0 MHz (-36.0 dB), 5334.0 MHz (-35.8 dB)
Data points	2402
Noise (σ, linear)	0.0020
Ripple (peak-to-peak)	–
S11 uncertainty	Calibration not confirmed – the data may be uncorrected; no uncertainty can be stated. Treat margins as unknown.

CALIBRATION AND TEST SETUP

Status	Not verified
Instrument-reported calibration	SOLT_1 (simulated)
Reference plane	Unknown
Calibration span	unknown
Calibration date / age	unknown
Calibration kit	–
Verification	not performed (50 Ω load check available on the Setup page)
Note	Calibration not confirmed by the user.
Note	Reference plane unknown.
Operator / project	– / –
DUT reference	–
Environment	not stated
Antenna	Dual-band 2.4/5 GHz whip (simulated) · type – · connector SMA
Cable / pigtail	–
Mounting / variant	– / –
Active GNSS antenna	no
Sweep plan	dualband: 2.4 GHz detail: 2000.000–3000.000 MHz, 1001 pts; 5 GHz detail: 4600.000–6000.000 MHz, 1401 pts
IFBW / averages / power	1000 Hz / 1 / -20 dBm
Limitation	Frequency suitability does not replace certification of the antenna with the radio module.
Limitation	Segment '5 GHz detail' limited to 4600.0–6000.0 MHz (requested 4600.0–6500.0 MHz; instrument limit). No statement is made beyond it.

GAIN AND RADIATION

Gain: Not measured. Radiation pattern: Not measured. A good match (S11) is necessary but not sufficient for good radiation: an antenna that dissipates energy in losses can show an excellent S11.

LIMITATIONS

• S11 describes only the impedance match at the reference plane. It does not demonstrate gain, efficiency, polarization purity, range, throughput, or regulatory compliance.

• Results depend on cable routing, enclosure, ground plane, and surroundings; re-measure the antenna in the final device.

• Not a substitute for certification or conformity testing (e.g. RED, FCC).

• Hardware validation status of this software version:
https://github.com/AltinumTechnologies/antennafox/blob/main/docs/HARDWARE_VALIDATION.md

INSTRUMENT, SOFTWARE AND DATASET

Backend	simulation (SIMULATION)
Identification	AntennaFox,Simulator,SIM-0001,1.0
Host software / firmware / HW	– / sim / sim
Serial number	SIMULATION
Instrument limits	0.100–6000.0 MHz (simulation)
Measurement ID / created	sample-sim_dualband_whip / 2026-10-11T12:00:00+00:00 (2026-10-11 14:00 CEST)
Report generated	2026-10-12 10:02 CEST
Dataset	sim_dualband_whip.afox
Catalog	2026.10-1
Software	AntennaFox 1.0.0